Solving Multiobjective Feature Selection Problems in Classification via Problem Reformulation and Duplication Handling

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***Abstract*—Reducing the number of selected features and improving the classification performance are two major objectives in feature selection, which can be viewed as a multiobjective optimization problem. Multiobjective feature selection in classification has its unique characteristics, such as it has a strong preference for the classification performance over the number of selected features. Besides, solution duplication often appears in both the search and the objective spaces, which degenerates the diversity and results in the premature convergence of the population. To deal with the above issues, in this article, during the evolutionary training process, a multiobjective feature selection problem is reformulated and solved as a constrained multiobjective optimization problem, which adds a constraint on the classification performance for each solution (e.g., feature subset) according to the distribution of nondominated solutions, with the aim of selecting promising feature subsets that contain more informative and strongly relevant features, which are beneficial to improve the classification performance. Furthermore, based on the distribution of feature subsets in the objective space and their similarity in the search space, a duplication analysis and handling method is proposed to enhance the diversity of the population. Experimental results demonstrate that the proposed method outperforms six state-of-the-art algorithms and is computationally efficient on 18 classification datasets.**

***Index Terms*—Classification, constraint handling, duplication analysis, feature selection, multiobjective optimization.**

I. Introduction

With the prevalence of big data in many domains, the existence of massive features is usually high-dimensional and has noisy, irrelevant, and redundant information, which impedes the performance of many supervised, unsupervised, and semi-supervised machine learning tasks, such as classification, regression, and clustering [1], [2]. As a data preprocessing and dimensionality reduction technique, feature selection has gained increasing attention, which plays a vital role in promoting the quality of the feature set. Feature selection in classification aims to select informative features and remove irrelevant, redundant, and noisy features from the original feature set that are able to discriminate samples from different classes, which can improve the classification accuracy, increase computational efficiency, reduce memory storage, and build better generalization classification models [3].

Feature selection is a complicated combinatorial optimization problem with 2^D possible candidate feature subset combinations, where D is the total number of features, which means the search space for feature selection problems grows exponentially as the number of features increases. Another difficulty in feature selection is the sophisticated interactions among features. For instance, two relevant features that retain similar information to the class label, can result in redundancy when they are selected together [4]. By contrast, two weakly relevant features that involve little information about the class label, can be complementary features. When they are selected together, the classification performance can be improved.

Over the past decades, many feature selection methods have been proposed, which can be broadly grouped as filter, wrapper, and embedded methods [3]. Filter methods calculate the score of each feature based on the intrinsic characteristics of the data and then select the discriminative features. Embedded methods usually perform feature selection and train a classifier simultaneously. Both filter and embedded methods are generally computationally efficient, but the classification performance is normally not as good as wrapper methods. Wrapper methods typically perform two steps: 1) a feature subset can be generated by a stochastic search or sequential search method and 2) a certain classifier acts as a black box for evaluating the quality of the candidate feature subsets, and the feature subset with the highest classification performance is returned as the selected features. Since sequential search tends to fall into local optimum, stochastic search is recently a more common strategy, of which the typical representative is evolutionary algorithms (EAs) [5]. EAs are inspired by biological evolution [6]. Benefiting from its numerous

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advantages, such as global search capability, no requirement for domain knowledge, and the population-based search can obtain multiple solutions, EAs have become promising tools for solving various complex optimization problems [7].

Minimizing the feature subset size and maximizing the classification performance are two key objectives in feature selection, which can be viewed as a multiobjective optimization problem. In single-objective feature selection, only the classification performance is adopted as the fitness function (e.g., [8], [9], and [10]), or the classification performance and the number of selected features are aggregated to a scalar fitness function (e.g., [11] and [12]). However, there are two main weaknesses in the scalar fitness function. First, it is not trivial to set a suitable coefficient to aggregate the two objectives in feature selection to properly reflect the preference of the user, especially when there is no prior knowledge about the dataset to be used. Second, only a single solution can be obtained, from which little insight into the feature selection problem can be gained. By contrast, treating feature selection as a multiobjective optimization problem is advantageous: 1) a set of nondominated feature subsets can be obtained to meet different requirements in real-world applications [13], and 2) a deeper insight into the learning problem (e.g., feature selection) can be gained by analyzing the nondominated front composed of multiple nondominated solutions [14], [15].

Multiobjective EAs (MOEAs) have shown their competitiveness in obtaining a set of tradeoff solutions for solving multiobjective optimization problems [16], [17], including multiobjective feature selection problems [18], [19]. Although feature selection is a multiobjective optimization problem, the preferences between its two objectives are different. From the point of view of most decision makers, maximizing the classification performance is more important than minimizing the feature subset size. Furthermore, in terms of the difficulty of optimizing these two objectives, maximizing the classification performance is more challenging than minimizing the feature subset size, since minimizing the feature subset size can be easily achieved by directly reducing the number of selected features, while maximizing the classification performance needs to consider the relevance and redundancy of features as well as the complex interactions between features. The different preferences between objectives in feature selection can also be reflected in the commonly used single-objective feature selection scalar fitness function [11], [12], where the coefficient imposed on the classification performance is much larger than that of the number of selected features.

Since feature selection is a kind of discrete optimization problem, another issue in multiobjective feature selection is the frequently appeared duplication in the objective space, especially when there are many redundant features or few training instances in a dataset. The existence of redundant and interactive features results in multiple feature subsets that select the same number of features and achieve the same classification performance. A large number of duplicated solutions can lead to poor diversity and premature convergence of the population. Accordingly, effectively handling such duplications is necessary. However, it is very challenging to decide which duplicated feature subsets in the objective space should be removed or retained.[1]

The overall goal of this article is to develop a novel method for handling the different preferences between objectives and frequently appeared duplicated feature subsets in multiobjective feature selection. To achieve this goal, this article proposes an MOEA based on problem reformulation and duplication handling, termed PRDH. The main contributions of this article can be summarized as follows.

1) A duplication handling method is proposed based on the distribution of feature subsets in the objective space and their similarity in the search space, with the aim of removing duplicated solutions in the objective space that have high similarity to other solutions but poor classification performance, thereby promoting the diversity of the population in both the objective and search spaces.
2) A problem reformulation method is proposed to focus more on the classification performance. During the evolutionary process, it reformulates a multiobjective feature selection problem as a constrained multiobjective optimization problem, which adds a constraint on the classification performance for each solution according to the distribution of the nondominated solutions, to treat solutions with different classification performances differently.
3) A constraint-handling technique is employed to address the reformulated constrained multiobjective feature selection problem, to preferentially select feasible feature subsets that contain more informative and strongly relevant features, which are beneficial to improve the classification performance.

The remainder of this article is organized as follows. Section II reviews the related work about multiobjective feature selection. The details of the proposed PRDH algorithm are elaborated in Section III. Section V presents the experiment design. Section V reports the experimental results. Section VI provides some further analysis. Finally, Section VII concludes this article.

II. Background and Related Work

A. Multiobjective Optimization

A multiobjective optimization problem with m objectives can be expressed as follows:

$$\begin{aligned} \min\ & \boldsymbol{F}(\boldsymbol{x}) = (f_1(\boldsymbol{x}), f_2(\boldsymbol{x}), \ldots, f_m(\boldsymbol{x}))^T \\ \text{where}\ & \boldsymbol{x} = (x_1, \ldots, x_D)^T \in \Omega \end{aligned} \tag{1}$$

where Ω is the search space and D is the dimensionality of a solution $\boldsymbol{x}$.

[1]Note that there are some similarities between multimodal multiobjective feature selection [20], [21], [22] and solution duplication in the objective space. For example, they both allow an algorithm to hold more than one solution with the same classification performance and select the same number of but different features. However, they also have some differences. Multimodal multiobjective feature selection focuses more on finding multiple optimal feature subsets for users to make preferential decisions, while solution duplication handling in the objective space focuses more on filtering out some unpromising duplicated feature subsets to improve the diversity and prevent premature convergence of the population.

In multiobjective optimization, the Pareto dominance relationship is often used to compare the quality of two solutions.

Definition 1 (Pareto Dominance Relation): For any two objective vectors $\boldsymbol{F}(\boldsymbol{x}_a)$ and $\boldsymbol{F}(\boldsymbol{x}_b)$, $\boldsymbol{F}(\boldsymbol{x}_a)$ can be said to dominate $\boldsymbol{F}(\boldsymbol{x}_b)$ if and only if the following two conditions hold.

1) $f_i(\boldsymbol{x}_a) \le f_i(\boldsymbol{x}_b)\ \forall i \in \{1, \ldots, m\}$.
2) $f_i(\boldsymbol{x}_a) < f_i(\boldsymbol{x}_b), \exists i \in \{1, \ldots, m\}$.

B. Related Work About Multiobjective Feature Selection

Although there are some works that use MOEAs for filter and embedded feature selection (e.g., [23], [24], and [25]), since the focus of this article is on wrapper methods, this section reviews some works for wrapper multiobjective feature selection methods.

In [13], a multiobjective particle swarm optimization (PSO) is proposed, which extends the multiobjective PSO [26] to deal with multiobjective feature selection problems. The results have shown that it not only outperforms the single-objective feature selection methods, but also performs better than some canonical MOEAs. Recognizing its superior performance, more and more MOEA-based feature selection methods have been developed to solve multiobjective feature selection problems in recent years [27], [28], [29].

Because of the need to reduce the number of features, feature selection can be regarded as a sparsity-based multiobjective optimization problem, and some sparsity-based works have shown promising results in solving multiobjective feature selection problems [30], [31]. For SparseEA [30], in initialization, it first evaluates the relevance of each feature to the classifier, and selects features with higher relevance via a tournament selection operator. Then, the evaluated feature relevance is used to guide the offspring reproduction process. One drawback of this method is that if the total number of features is large, it will consume a large number of evaluations on initialization. In addition to the number of selected features and the classification error, in EvoFS [32], a third objective of the variance of the lowest-performing feature in the feature subset, is constructed to search for nonseparable optimal feature subsets.

In real-world applications, the datasets may contain noisy data. Focusing on classification datasets with missing data, in addition to the classification performance and the feature subset size, in [33], the missing rate is considered as the third objective which formulates a feature selection problem as a tri-objective optimization problem, to further improve the reliability of the selected features. In [34], the classification performance and the reliability degree of data are treated as a bi-objective optimization problem, to search for feature subsets with good classification performance and high reliability. Sometimes, the cost associated with each feature is not precise. To solve such problems, in [35], a fuzzy dominance relationship and a fuzzy crowding distance are defined to obtain feature subsets with high classification performance and low cost.

The two objectives in feature selection are not always in conflict with each other. For instance, removing irrelevant and noisy features can improve classification performance. A dynamic reference-based MOEA [36] is proposed to detect the conflicting regions and allocate more computational resources to the conflicting regions. The optimal feature subsets for a classification problem could be multiple. To search for these multiple optimal feature subsets, MOEAs are designed to consider not only the performance of solutions in the objective space, but also the distance between solutions in the search space [20], [21].

The search space of feature selection problems grows exponentially as the number of features increases. Improving the search efficiency of an EA can help to solve high-dimensional feature selection problems. For example, a multiobjective differential evolution algorithm [37] is proposed for feature selection. In this method, the probability difference-based binary mutation and the one-bit purifying search are embedded into the differential evolution algorithm to improve its search capabilities. A cut-point PSO algorithm [38] is proposed for discretization-based multiobjective feature selection, which can select an arbitrary number of cut-points for the discretization to better explore relevant features. The steering-matrix-based MOEA [39] intends to select important (relevant) features guided by the steering matrix, which is constantly updated in the process of evolution. Meanwhile, the steering matrix is also used to remove irrelevant features as dimensionality reduction. In the variable granularity search-based MOEA [40], with the evolution process, the search granularity changes from coarse to fine to reduce the search space.

Recently, to solve the frequently appeared duplications in multiobjective feature selection, a duplication analysis EA (DAEA) [41] is proposed, which measures the distance between every two solutions in the population as their similarities, and those with high similarities tend to be removed during the environmental selection procedure to enhance the population diversity.

C. Discussions

Different methods have different focuses. Table I summarizes some existing wrapper-based multiobjective feature selection algorithms. The work in [41] has shown effectiveness in handling the frequently appeared duplication in multiobjective feature selection. However, by taking a closer look, we can find that it exhibits a high computational complexity with low efficiency (this will be verified in Section V-D). This is because the distance between every two solutions in the population needs to be calculated, which prevents it from scaling well to high-dimensional datasets.

Another issue is the different preferences between objectives in multiobjective feature selection. It is known that the commonly used two objectives (i.e., classification performance and the number of selected features) have different levels of importance. Most current works treat these two objectives equally important [28], which is not true in real-world applications. Zhou et al. [42] defined an accuracy-preferred dominance relationship, where a solution is better than the other, not only based on the canonical Pareto dominance relationship, but also with the classification accuracy is higher than a predefined

TABLE I
SUMMARY OF WRAPPER-BASED MULTIOBJECTIVE FEATURE SELECTION ALGORITHMS

Algorithm	Objectives	Different objective preferences?	Duplication handling (objective space)?
CMDPSOFS [13]	Classification performance, feature subset size	×	×
NSGA-III [33]	Classification performance, feature subset size, missing rate	×	×
BMOPSOFS [34]	Classification performance, feature subset size	×	×
PSOMOFS [35]	Classification performance, fuzzy cost	×	×
MOEA/D-DYN [36]	Classification performance, feature subset size	✓	×
MOFS-BDE [37]	Classification performance, feature subset size	×	×
FCPSO [38]	Classification performance, feature subset size, discriminate distance	×	×
SM-MOEA [39]	Classification performance, feature subset size	×	×
VGS-MOEA [40]	Classification performance, feature subset size	×	×
PS-NSGA [42]	Classification performance, feature subset size	✓	×
PMOFS [43]	Classification performance, feature subset size	✓	×
EvoFS [32]	Classification performance, feature subset size, variance of the lowest-performing feature	×	✓
SIOM-NSGA-II [44]	Classification performance, feature subset size	×	✓
SparseEA [30]	Classification performance, feature subset size	×	×
PMMOEA [31]	Classification performance, feature subset size	×	×
DAEA [41]	Classification performance, feature subset size	×	✓
PRDH	Classification performance, feature subset size	✓	✓

tolerance, in order to make solutions with higher classification accuracy more likely to survive. However, setting an appropriate tolerance manually is not an easy job since it is dataset dependent. In [43], we propose a preference-inspired MOEA to put more emphasis on classification performance, which is achieved by dynamically adjusting the different selection pressures of the two objectives. Nevertheless, dynamically adjusted selection pressure cannot adapt to the population distribution in different states.

In view of the limitations of previous work, in the next section, first, a duplication analysis and handling method is proposed to filter out the unpromising duplicated solutions in the objective space. The proposed duplication handling method has a low computational cost and high efficiency, since it only considers the distance (similarity) between each duplicated solution and its nearest nondominated solution. Then, a problem reformulation method and a constraint handling method are proposed to put more emphasis on classification performance. Unlike most of the current work on constrained multiobjective optimization, which aims to solve static constraints, i.e., the constraint functions have been defined before solving the problem and remain unchanged during the optimization process [45], [46], the work of this article aims to impose and handle dynamic constraints, i.e., the constraint functions might change in each generation according to the distribution of the nondominated feature subsets.

Note that in [47], a feature selection problem is also formulated as a constrained optimization problem, where a predefined maximum number of selected features is taken as a constraint. However, different from [47] that focuses on solving single-objective feature selection problems, the constraint in this work is imposed on the classification performance in the process of problem-solving according to the distribution of nondominated solutions, with the aim of obtaining a set of tradeoff solutions that have better classification performance than the nondominated solution that has the smallest number of selected features.

III. PROPOSED METHOD

This section first describes the problem formulation of the multiobjective feature selection to be solved, then presents the overall framework of the proposed PRDH method, and finally elaborates on each main component of PRDH.

A. Problem Statement: Multiobjective Feature Selection

A multiobjective feature selection problem can be expressed as follows:

$$\begin{aligned} \min\ & \boldsymbol{F}(\boldsymbol{x}) = (f_{\text{ratio}}(\boldsymbol{x}), f_{\text{err}}(\boldsymbol{x}))^T \\ \text{where}\ & \boldsymbol{x} = (x_1, \ldots, x_D)^T \in \{0, 1\}^D \end{aligned} \tag{2}$$

where D is the total number of features in a dataset. $\boldsymbol{x}$ is the solution that represents a feature subset. In this article, binary encoding is used since it is more straightforward than a continuous encoding in feature selection [5], [28], where $\boldsymbol{x} \in \{0, 1\}^D$ is a bit-string representation. $x_i = 1$ and $x_i = 0$ denote that the ith feature is selected and discarded, respectively.

The first objective $f_{\text{ratio}}(\boldsymbol{x})$ is the selected feature ratio, which equals the ratio of the number of selected features to the total number of features

$$f_{\text{ratio}}(\boldsymbol{x}) = \frac{\sum_{i=1}^{D} x_i}{D}. \tag{3}$$

The second objective $f_{\text{err}}(\boldsymbol{x})$ is the classification error rate, which is related to classification performance and can be obtained by evaluating a certain classifier using the selected feature subset $\boldsymbol{x}$. Since many collected data are unbalanced, a balanced classification error rate [48] is used in this article

$$f_{\text{err}}(\boldsymbol{x}) = 1 - \frac{1}{c} \sum_{i=1}^{c} \frac{TP_i}{|S_i|} \tag{4}$$

where c is the number of classes of a dataset. TP_i represents the number of correctly identified instances in class i, and $|S_i|$ denotes the sample size of class i. Since the balanced accuracy has no bias to any class in the classification problem, the weight for each class is set to $1/c$.

B. Overall Framework of PRDH

The overall framework of PRDH is given in Fig. 1. PRDH first generates an initial population $\mathcal{P}$ with N solutions (e.g.,

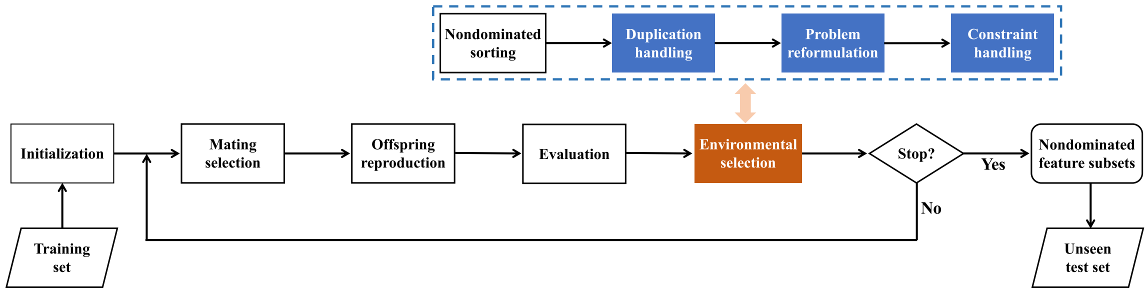

Fig. 1. Flowchart of PRDH.

feature subsets). The number of selected features in each initial solution $\boldsymbol{x}$ is randomly generated, i.e., $|\boldsymbol{x}| = \text{rand}(1, D)$. However, when the total number of features D is large, as suggested by [41], we set the number of selected features to be no more than three times the population size N, i.e., $|\boldsymbol{x}| = \text{rand}(1, \min(D, 3N))$, which facilitates convergence and dimensionality reduction for high-dimensional datasets.

Like most of the canonical MOEAs, each generation in the main loop of PRDH consists of four operations: 1) mating selection; 2) offspring reproduction; 3) evaluation; and 4) environmental selection. Mating selection is performed to select good solutions from the parent population $\mathcal{P}$ to generate offspring solutions. Subsequently, the offspring reproduction operators, i.e., single-point crossover and bit-flip mutation, can be utilized to produce an offspring population $\mathcal{O}$. After evaluating the classification performance of each solution in $\mathcal{O}$, the environmental selection operator is performed to choose elite solutions to survive from the union of the parent and offspring populations (i.e., $\mathcal{P}\bigcup\mathcal{O}$). The above evolutionary learning procedure repeats until the stopping criterion is met. Finally, the nondominated feature subsets are selected from the final population as the output.

The main contributions in PRDH lie in the three other components in the environmental selection, i.e., duplication handling (Algorithm 1), problem reformulation (6), and constraint handling (Algorithm 2), which will be described in detail below.

C. Duplication Handling

Duplicated solutions can be in two types: 1) duplicated solutions in the search space and 2) duplicated solutions in the objective space. Duplicated solutions in the search space represent that the feature subsets select the same features. These duplicated solutions in the search space can be easily identified and removed before classification performance evaluation to save computational cost. By contrast, solutions duplicated only in the objective space indicate that feature subsets select the same number of but different features, and these feature subsets have the same classification performance. They are more difficult to handle than those duplicated solutions in the search space. In this section, we focus on dealing with solution duplication in the objective space.

Fig. 2 gives an example to facilitate the understanding of the proposed duplication handling method. Unlike in continuous optimization problems, where the population may be continuously distributed in the objective space, in feature selection problems, the distribution of solutions in the objective space is discrete. Suppose solutions $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ are two duplicated solutions in the objective space, whereas they both select three features and produce the same classification performance. Now we give an analysis of whether they should be deleted or reserved.

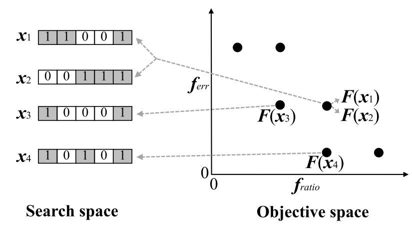

Fig. 2. Example of duplication analysis.

We first find the reference point $\mathcal{R}_e$ of $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ in terms of the objective of classification error rate, which is a nondominated solution that has the same or most similar classification error rate with $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ (line 5 of Algorithm 1), i.e., $\boldsymbol{x}_3$ in Fig. 2. $\boldsymbol{x}_3$ selects a smaller number of features than $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ but exhibits the same classification performance with $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$. To measure the similarity between $\boldsymbol{x}_3$ and $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ in the search space, respectively, we find the position of each selected feature in $\boldsymbol{x}_3$, i.e., the first and fifth features. Then, the Hamming distance between $\boldsymbol{x}_3$ and the duplicated solutions $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ at positions where the bit of $\boldsymbol{x}_3$ is 1 are calculated as the similarity between them. The Hamming distance measures the ratio of positions at which the two feature subsets are different, i.e., $d_e(\boldsymbol{x}_1, \boldsymbol{x}_3) = 0$ and $d_e(\boldsymbol{x}_2, \boldsymbol{x}_3) = 0.5$ if only the first and fifth features are considered. The Hamming distance $d_e(\boldsymbol{x}_1, \boldsymbol{x}_3)$ equals 0 implying that all features selected by $\boldsymbol{x}_3$ are also selected by $\boldsymbol{x}_1$. Although $\boldsymbol{x}_1$ selects some more features (e.g., the second feature in $\boldsymbol{x}_1$) than $\boldsymbol{x}_3$, these extra features selected by $\boldsymbol{x}_1$ do not improve the classification performance, which suggests that these extra features are likely to be redundant or irrelevant. The smaller the distance d_e, the more common features they select, which suggests the more irrelevant or redundant features that a duplicated solution might has. From this perspective, we prefer to reserve duplicated solutions that have a larger distance d_e with their reference point $\mathcal{R}_e$.

We continue to analyze the duplication in terms of the objective of a number of selected features. In the similar

Algorithm 1: Duplication Handling

Input: $\mathcal{U}$: the union of the parent population $\mathcal{P}$ and the offspring population $\mathcal{O}$;
$\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_l$: the nondominated front level for each solution in $\mathcal{U}$.
Output: the population $\mathcal{U}$.

```
1  U = {x | x ∈ F_1};
2  for i = 2 : l do
3      Find all duplicated solutions (in the objective space) in F_i;
4      for each duplicated solution x ∈ F_i do
5          R_e ← {x' | argmin_{x'∈F_1} (|f_err(x') − f_err(x)|)};
6          d_e(x, R_e) ← Calculate the Hamming distance between x and R_e at the positions where the bit of R_e is 1.;
7          R_r ← {x' | argmin_{x'∈F_1} (|f_ratio(x') − f_ratio(x)|)};
8          d_r(x, R_r) ← Calculate the Hamming distance between x and R_r;
9          Save nondominated duplicated solutions to U based on Eq. (5).
10 Return U.
```

Algorithm 2: Constraint Handling

Input: $\mathcal{U}$: the union of the parent population $\mathcal{P}$ and the offspring population $\mathcal{O}$ after duplication handling;
N: population size.
Output: the new parent population $\mathcal{P}$.

```
Divide U into a feasible set S_fe and an infeasible set S_infe;   /* Eqs. (7)-(8) */;
P = ∅;
if |S_fe| > N then
    Select best N feature subsets out of S_fe based on nondominated front levels and crowding distance values;
    Put the chosen N feature subsets into P;
else
    P ← S_fe;
    Sort feature subsets in S_infe in light of their degree of constraint violation (max{0, f_err(x) − f_err(x_t)}), and put the best (N-|S_fe|) feature subsets in the sorted S_infe into P;
Return P.
```

way, we find the nondominated solution that selects the same or similar number of features with $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ as their reference point $\mathcal{R}_r$ (line 7 of Algorithm 1), i.e., $\boldsymbol{x}_4$ in Fig. 2. $\boldsymbol{x}_4$ selects the same number of features with $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ but exhibits the better classification performance than that of $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$. The Hamming distance between $\boldsymbol{x}_4$ and the duplicated solutions $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ are calculated, respectively, i.e., $d_r(\boldsymbol{x}_1, \boldsymbol{x}_4) = 0.33$ and $d_r(\boldsymbol{x}_2, \boldsymbol{x}_4) = 0.33$. The smaller the distance d_r, the more common features of the duplicated solutions (i.e., $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$), and their reference point $\mathcal{R}_r$ (i.e., $\boldsymbol{x}_4$) selection. Since the duplicated solutions have worse classification performance than their reference point $\mathcal{R}_r$, from the perspective of diversity and relevant and redundancy analysis, among these duplicated solutions, we prefer to reserve the solution with a large distance d_r from their reference point $\mathcal{R}_r$.

Algorithm 1 gives the details of the proposed duplication analysis and handling method, which aims to remove unpromising duplicated solutions in the objective space that contain more irrelevant and redundant features. In Algorithm 1, after performing the nondominated sorting operator, the duplicated solutions in the objective space for each nondominated front level are first found. For each duplicated solution, we find its reference points $\mathcal{R}_e$ and $\mathcal{R}_r$ according to its two objective values, i.e., f_{err} and f_{ratio}. Note that R_e and R_r are nondominated solutions who have the smallest objective value with that of $\boldsymbol{x}$ according to f_{err} and f_{ratio} (lines 5 and 7 of Algorithm 1), respectively. Once R_e and R_r are determined for each duplicated solution $\boldsymbol{x}$, the similarities between $\boldsymbol{x}$ and its reference points R_e and R_r, i.e., Hamming distance, are calculated.

Intuitively, we should remove the duplicated solutions in the objective space with smaller distances (d_e and d_r), because the smaller distances indicate that this solution selects more common features with its reference points R_e and R_r, respectively. One duplicated solution may have a larger d_e and a smaller d_r, while the other one may have a smaller d_e and a larger d_r. Therefore, it is hard to decide which duplicated solution to remove. In this article, we propose using the Pareto dominance relationship (Definition 1) to decide which duplicated solutions should be deleted or saved. Specifically, we formulate the similarities of a duplicated solution $\boldsymbol{x}$ with its two reference points $\mathcal{R}_e$ and $\mathcal{R}_r$ as a new (maximization) bi-objective optimization problem as follows:

$$\max \; (d_e(\boldsymbol{x}, \mathcal{R}_e), d_r(\boldsymbol{x}, \mathcal{R}_r))^T. \tag{5}$$

Only those duplicated solutions in the objective space that are not dominated by any other solutions are retained in the population. For example, in Fig. 2, among the two duplicated solutions in the objective space $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$, $\boldsymbol{x}_1$ will be deleted while $\boldsymbol{x}_2$ will be retained since $\boldsymbol{x}_2$ can dominate $\boldsymbol{x}_1$ according to (5). In this way, a number of duplicated solutions in the objective space that have high similarities and contain more irrelevant and redundant features can be removed from the population.

D. Problem Reformulation

After deleting some unpromising duplicated solutions in the objective space via the proposed duplication handling method, we give an analysis on the distribution of the remaining solutions. Fig. 3 gives an example of the distribution of solutions in the objective space. In Fig. 3(a), we regard $\boldsymbol{x}_t$ as a tipping point, which represents the solution that has the smallest selected feature ratio value among all nondominated solutions, i.e., $\boldsymbol{x}_t = \arg\min_{\boldsymbol{x} \in \mathcal{F}_1} f_{\text{ratio}}(\boldsymbol{x})$, where $\mathcal{F}_1$ is the first nondominated front level after the nondominated sorting operator [49].

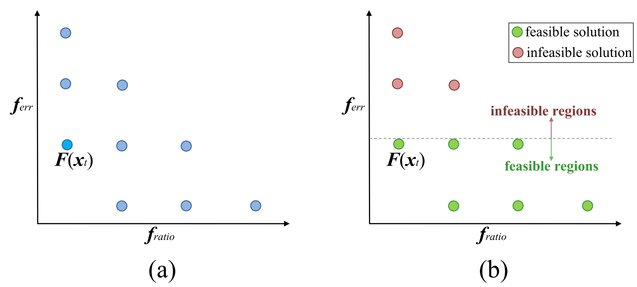

Fig. 3. Schematic of reformulating a multiobjective feature selection problem as a constrained multiobjective optimization problem. (a) Distribution of the population. (b) Divide the population into feasible and infeasible solutions.

Among all the nondominated solutions, the tipping point $\boldsymbol{x}_t$ selects the minimal number of features but achieves the worst classification performance.

Based on the relationship between the classification error rate of each solution in the population and that of the tipping point $\boldsymbol{x}_t$, the solutions can be divided into two groups, i.e., the red points and the green points in Fig. 3(b). The red points have relatively large classification error rates while the green points exhibit relatively small classification error rates but select a relatively large number of features. Depending on the relationship with the tipping point $\boldsymbol{x}_t$ in terms of the number of selected features, the red points can be further divided into two types.

1) The red points select the same number of features as that of the tipping point $\boldsymbol{x}_t$, but have larger classification error rates than that of $\boldsymbol{x}_t$. This means that compared with $\boldsymbol{x}_t$, these red points contain more irrelevant features or the combinations of these selected features are not good.
2) The red points select more features than that of the tipping point $\boldsymbol{x}_t$ and even have larger classification error rates, which suggests there are many irrelevant or redundant features in these red points that deteriorate their classification performance.

Although these red points select relatively small feature subsets than some of the green points, they (red points) contribute little to improving the classification performance, since they select fewer relevant or more redundant features and even exhibit larger classification error rates than that of the tipping point $\boldsymbol{x}_t$. In feature selection, the classification performance is more important than the number of selected features. Meanwhile, maximizing the classification performance (or minimizing the classification error rate) is more challenging than minimizing the feature size, since minimizing the feature size can be easily achieved by directly reducing the number of selected features, while maximizing the classification performance needs to consider the complex combinations of different features. It is of great significance to prioritize solutions with higher classification performance (smaller classification error rates).

To this end, during the evolutionary training process, based on the relationship between the classification error rate value of solutions in the population and the classification error rate value of the tipping point, we reformulate a multiobjective feature selection problem as a constrained multiobjective optimization problem, which adds a constraint on the classification performance for each solution. The reformulated constrained multiobjective feature selection problem has the form of

$$\begin{aligned} \min\ & \boldsymbol{F}(\boldsymbol{x}) = (f_{\text{ratio}}(\boldsymbol{x}), f_{\text{err}}(\boldsymbol{x}))^T \\ \text{s.t.}\ & g(\boldsymbol{x}) = f_{\text{err}}(\boldsymbol{x}) - f_{\text{err}}(\boldsymbol{x}_t) \le 0 \\ \text{where}\ & \boldsymbol{x}_t = \arg\min_{\boldsymbol{x}\in\mathcal{F}_1} f_{\text{ratio}}(\boldsymbol{x}) \\ & \boldsymbol{x} = (x_1, \ldots, x_D)^T \in \{0, 1\}^D \end{aligned} \tag{6}$$

where $\boldsymbol{x}_t$ is the tipping point, which represents the solution that has the smallest selected feature ratio value among all nondominated solutions. The inequality constraint $g(\boldsymbol{x})$ means that if the classification error rate of a solution $\boldsymbol{x}$ is smaller than that of the tipping point $\boldsymbol{x}_t$, this solution is a feasible one ($g(\boldsymbol{x}) \le 0$); otherwise, it is an infeasible one ($g(\boldsymbol{x}) > 0$).

The aim of (6) is to strive to distinguish solutions with poor classification performance that contain more irrelevant and redundant features, thus getting solutions focused on regions with better classification performance, avoiding wasting computational costs in unpromising regions. It is worth noting that the tipping point $\boldsymbol{x}_t$ in (6) can change in each generation, since the distribution of the population in the objective space is different in each generation, and the corresponding nondominated solutions will also change. During the evolutionary training process, as the tipping point $\boldsymbol{x}_t$ with a smaller classification error rate is found, the range of the feasible region is also reduced.

E. Constraint Handling

The key idea of constraint handling is to retain more feasible solutions, which contain more informative and strongly relevant features, while avoiding the selection of infeasible solutions that contain many irrelevant features.

At each generation, after problem reformulation, based on the feasibility of each feature subset in (6), the combination of the parent population $\mathcal{P}$ and the offspring population $\mathcal{O}$ can be separated into a feasible set

$$\mathcal{S}_{fe} = \left\{\boldsymbol{x} \in \mathcal{P} \bigcup \mathcal{O} | g(\boldsymbol{x}) \le 0\right\} \tag{7}$$

and an infeasible set

$$\mathcal{S}_{\text{infe}} = \left\{\boldsymbol{x} \in \mathcal{P} \bigcup \mathcal{O} | g(\boldsymbol{x}) > 0\right\}. \tag{8}$$

The solutions in the feasible solution set $\mathcal{S}_{fe}$ have relatively small classification error rates, while those in the infeasible solution set $\mathcal{S}_{\text{infe}}$ have relatively large classification error rates. The detailed process of handling constraints is illustrated in Algorithm 2. The aim of constraint handling is to obtain a set of tradeoff feature subsets, which achieve better classification performance than that of the tipping point $\boldsymbol{x}_t$. To achieve this, we prefer feasible solutions to infeasible ones and give a priority to feasible solutions during the environmental selection process. Based on the relationship between the feasible set size and the population size N, the selection of elite solutions from the union of the parent and offspring populations as the next parent population can be divided into the following two cases.

1) If $|\mathcal{S}_{fe}| > N$, which means there are enough feasible solutions with smaller classification error rates to be selected as the next parent population. In this case, the nondominated front levels and crowding distance values are used as the criteria to select the best N solutions out of $\mathcal{S}_{fe}$. The selected N solutions are then used as the next parent population $\mathcal{P}$.
2) If $|\mathcal{S}_{fe}| \leq N$, which suggests that the number of feasible solutions is smaller than the population size N. Under this situation, all the feasible solutions in $\mathcal{S}_{fe}$ are directly added to the next parent population $\mathcal{P}$. The rest (N–$|\mathcal{S}_{fe}|$) solutions are chosen from $\mathcal{S}_{\text{infe}}$. Since we want to save more feature subsets with smaller classification error rates, herein, we sort the solutions in $\mathcal{S}_{\text{infe}}$ in an ascending order according to their degree of constraint violation $G(\boldsymbol{x})$ (i.e., $G(\boldsymbol{x}) = \max\{0, f_{\text{err}}(\boldsymbol{x}) - f_{\text{err}}(\boldsymbol{x}_t)\}$), and then put the top (N–$|\mathcal{S}_{fe}|$) solutions in the sorted $\mathcal{S}_{\text{infe}}$ into $\mathcal{P}$.

Through the above selection process, solutions with good convergence (e.g., nondominated solutions) and small classification error rates are prioritized, which contributes to producing a set of feature subsets with better classification performance.

F. Analysis of Computational Complexity

The classification performance of each feature subset is affected by the classifier it uses. In addition to the classification performance evaluation, the computational complexity of PRDH is mainly governed by three operators: 1) mating selection; 2) offspring reproduction; and 3) environmental selection. The worst-case time complexities of mating selection and offspring reproduction are $O(N)$ and $O(DN)$, respectively, where N is the population size and D is the total number of features in a dataset. The environmental selection operation consists of four steps: 1) nondominated sorting; 2) duplication handling; 3) problem reformulation; and 4) constraint handling. The nondominated sorting operator has a computational complexity of $O(N^2)$. The time complexity of the duplication handling operator is about $O(nD)$, where n is the total number of duplicated solutions and $n \ll N$. Finally, the problem reformulation and constraint-handling operators have a computational complexity of $O(N \log N)$.

To summarize, the overall complexity of one generation of PRDH is bounded by $O(N^2)$ or $O(DN)\}$, whichever is larger.

IV. EXPERIMENT DESIGN

A. Classification Datasets

A total of 18 classification datasets collected from the UCI machine learning repository [50] are utilized for testing the performance of the proposed PRDH algorithm and its competitors. These datasets cover a variety of data types. The number of features, classes, and instances of these 18 datasets have a large range of variations, which can be more convincing to test the performance of each compared algorithm. Table II gives a summary of these datasets.

TABLE II
CHARACTERISTICS OF THE CLASSIFICATION DATASETS

Dataset	#Features (D)	#Classes (c)	#Instances
Vehicle	18	4	846
German	24	2	1000
Ionosphere	34	2	351
Sonar	60	2	208
Mice	77	8	1077
Movementlibras	90	15	360
Hillvalley	100	2	1212
Musk1	166	2	476
Semeion	265	2	1593
Arrhythmia	278	13	452
LSVT	310	2	126
Madelon	500	2	2600
Gametes	1000	2	1600
ORL	1024	40	400
SRBCT	2308	4	83
DLBCL	5469	2	77
Prostate-GE	5966	2	102
ALLAML	7129	2	72

B. Comparison Algorithms

Six MOEAs are selected as competitors of PRDH: 1) NSGA-II [49]; 2) MOEA/D [51]; 3) SIOM-NSGA-II [44]; 4) SparseEA [30]; 5) PMMOEA [31]; and 6) DAEA [41]. The reasons for choosing the above six MOEAs as competitors are: NSGA-II and MOEA/D are representatives of Pareto dominance-based and decomposition-based MOEAs, while SIOM-NSGA-II, SparseEA, PMMOEA, and DAEA represent state-of-the-art MOEA-based methods that are specifically designed for feature selection.

C. Experimental Settings

The experiments for MOEAs are conducted on the open-source platform PlatEMO [52]. Each MOEA independently runs 30 times. The population size for all algorithms is set to be the same as the number of features ($N = D$). Nevertheless, to avoid high computational costs on high-dimensional datasets, the population size is bounded by 200 ($N = 200$ if $D > 200$) [44]. For a fair comparison, NSGA-II, MOEA/D, and the proposed PRDH algorithm all use the single-point crossover and bit-flip mutation operators to generate offspring solutions, where the crossover probability and mutation probability are set to 1.0 and $1/D$, respectively, while SparseEA, SIOM-NSGA-II, PMMOEA, and DAEA adopt their specifically designed genetic operators. The customizable parameters for each compared MOEA follow their original papers, which have shown their effectiveness.

1) *MOEA/D [51]:* The number of the weight vectors in the neighborhood of each weight vector is set to $T = 0.2N$, where N is the population size; the maximal number of solutions replaced by each offspring solution is $n_r = 2$; the probability that parent solutions are selected from the neighborhood is set to $\delta = 0.9$.
2) *SIOM-NSGA-II [44]:* The initial population is divided into three subpopulations, and the features of these three subpopulations are selected based on selection probabilities of approximately 25%, 50%, and 75%, respectively.
3) *PMMOEA [31]:* The population size and the number of generations for pattern mining are set to $0.2N$ and 10, respectively.

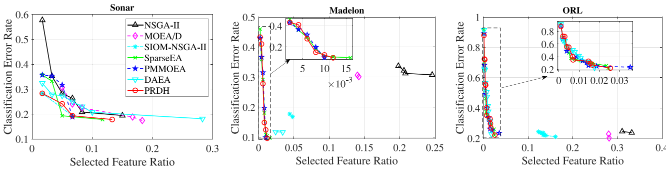

Fig. 4. Distributions of nondominated feature subsets obtained by each algorithm on test sets in terms of median HV value.

4) *DAEA [41]:* The neighborhood size $T = \max(4, 0.2N)$); the local mating and global mating rates are set to 0.8 and 0.2, respectively; and the modified mutation ratio is 0.8.

Following [11] and [27], each classification dataset is randomly partitioned into a training data subset and a test data subset with ratios of 70% and 30%, respectively. During the training process, to avoid the feature selection bias, K-NN with five-fold cross-validation on the training data subset is used to calculate the classification error rate [53]. The reason for choosing K-NN as the classifier is that, compared with other classifiers, K-NN has the following advantages: it is easy to implement and intuitive to understand; it can learn nonlinear decision boundaries when used for classification; last but not least, the classification performance of K-NN can severely deteriorate by the presence of noisy or irrelevant features, so removing noisy and irrelevant features by feature selection can greatly improve the classification performance. In K-NN, an even value of K can lead to a tie in samples at the decision boundaries, so it is suggested to use an odd number for K. Additionally, a smaller value of K (e.g., $K = 1$) is more affected by noise on the results, which results in the low classification accuracy. By contrast, a larger value of K will have smoother decision boundaries which means lower variance but increased bias, and also a high computational cost, which leads to the low efficiency. In this article, we set $K = 5$ for the balance between accuracy and efficiency [11], [44]. The nondominated feature subsets obtained at the end of the training process are used for the unseen test data subset to obtain and report the final classification error rates.

D. Performance Metrics

A feature selection method can be assessed by two aspects: 1) effectiveness and 2) efficiency. The following five performance metrics are used to measure the effectiveness or efficiency of each compared method.

1) *Hypervolume (HV) [54]:* It measures both the convergence and diversity of nondominated feature subsets.
2) *Inverted Generational Distance (IGD) [55]:* It measures both the convergence and diversity of nondominated feature subsets. Since the true Pareto front of feature selection in each dataset is usually unknown, to calculate IGD, the nondominated solutions in the union population of all compared MOEAs are taken as the true Pareto front.
3) *Minimal Classification Error Rate (MCER):* Among the multiple nondominated solutions/feature subsets, the feature subset with the smallest training classification error rate is chosen and its testing classification error rate is recorded as MCER.
4) *Number of Selected Features (FN):* It represents the number of selected features by the feature subset with the minimum classification error rate.
5) *Training Time:* It measures the efficiency of each compared algorithm on the training data.

V. Results and Discussions

In this section, we attempt to answer the following two questions by assessing the performance of the proposed method: 1) What is the quality of selected features by PRDH? 2) What is the efficiency of the proposed PRDH compared with the benchmark methods?

A. Comparisons in Terms of the Multiobjective Performance Metrics

Table III lists the average HV results obtained by all the competitor algorithms. The larger the HV value of an algorithm, the better its performance. As indicated in the table, when measured by HV, the proposed PRDH method performs best on 11 out of the 18 datasets. It can also be observed that the advantage of PRDH is more obvious on the medium and high-dimensional classification datasets, which are more challenging to solve. In terms of Wilcoxon's rank-sum test results, the proposed PRDH method performs significantly better than NSGA-II, MOEA/D, SIOM-NSGA-II, SparseEA, and DAEA on 15, 17, 10, 5, 4, and 2 out of 18 datasets. By contrast, these six competitors are not significantly superior to PRDH in any dataset. Friedman's test results in Table III also show that PRDH ranks first among these seven MOEAs, which suggests it has the best performance according to HV.

For intuitive analysis, Fig. 4 plots the distribution of nondominated feature subsets on the test data obtained by each algorithm. Note that the results are from the run producing the median HV values of the 30 runs. The results are presented on the Sonar, Madelon, and ORL datasets, respectively, which represent the results on the small, medium, and large datasets. Similar patterns can also be observed on the other datasets. From Fig. 4, we can see that on the Sonar dataset, thanks to the proposed problem reformulation technique that focuses more on the classification performance, the nondominated feature subsets obtained by PRDH tend to have smaller classification error rates, since they can dominate most feature subsets obtained by other algorithms. On the medium and large datasets Madelon and ORL, the feature subsets obtained by NSGA-II, MOEA/D, and SIOM-NSGA-II are gathered in a small region in the objective space and have a large number of selected features. By contrast, those obtained by SparseEA, PMMOEA, DAEA, and the proposed PRDH method have a good distribution in terms of the classification error rate and the number of selected features. Among these four MOEAs, the feature subsets obtained by PRDH are very competitive in terms of both the classification performance and the solution distributions, since the proposed problem reformulation technique pays more attention to the objective of

TABLE III
COMPARATIVE RESULTS OF **HV** VALUES (MEAN AND STANDARD DEVIATION) OBTAINED BY NSGA-II, MOEA/D, SIOM-NSGA-II, SPARSEEA, PMMOEA, DAEA, AND PRDH ON THE TEST DATA

Dataset	**NSGA-II** [49]	**MOEA/D** [51]	**SIOM-NSGA-II** [44]	**SparseEA** [30]	**PMMOEA** [31]	**DAEA** [41]	**PRDH**
Vehicle	6.79e-1(1.50e-2)≈	6.57e-1(1.98e-2)+	6.80e-1(1.53e-2)≈	6.79e-1(1.21e-2)≈	6.79e-1(1.26e-2)≈	6.79e-1(1.30e-2)≈	6.77e-1(1.31e-2)
German	6.59e-1(6.33e-2)≈	6.49e-1(5.08e-2)+	6.52e-1(5.24e-2)+	6.62e-1(6.15e-2)≈	6.73e-1(2.71e-2)≈	6.72e-1(5.03e-2)≈	6.75e-1(4.62e-2)
Ionosphere	9.12e-1(1.89e-2)≈	9.04e-1(2.04e-2)≈	9.09e-1(2.16e-2)≈	9.14e-1(2.09e-2)≈	9.05e-1(1.95e-2)≈	9.16e-1(1.78e-2)≈	9.11e-1(1.92e-2)
Sonar	8.03e-1(3.13e-2)+	8.01e-1(2.55e-2)+	8.15e-1(2.57e-2)≈	8.23e-1(2.13e-2)≈	8.12e-1(2.54e-2)≈	8.16e-1(1.70e-2)≈	8.19e-1(2.18e-2)
Mice	9.39e-1(1.33e-2)+	9.22e-1(8.82e-3)+	9.55e-1(8.84e-3)≈	9.58e-1(6.10e-3)≈	9.56e-1(6.58e-3)≈	9.57e-1(7.34e-3)≈	9.54e-1(9.55e-3)
Movementlibras	7.32e-1(3.48e-2)+	7.32e-1(2.87e-2)+	7.50e-1(3.10e-2)≈	7.64e-1(2.40e-2)≈	7.50e-1(2.71e-2)≈	7.54e-1(2.65e-2)≈	7.57e-1(3.13e-2)
Hillvalley	6.05e-1(2.30e-2)+	6.15e-1(1.62e-2)+	6.24e-1(1.30e-2)≈	6.21e-1(1.34e-2)≈	6.13e-1(1.80e-2)+	6.26e-1(1.38e-2)≈	6.27e-1(1.46e-2)
Musk1	8.12e-1(3.06e-2)+	8.17e-1(2.83e-2)+	8.62e-1(2.61e-2)≈	8.65e-1(2.25e-2)≈	8.48e-1(3.05e-2)≈	8.71e-1(2.30e-2)≈	8.63e-1(2.61e-2)
Semeion	8.43e-1(1.59e-2)+	8.95e-1(2.39e-2)+	9.38e-1(1.28e-2)+	9.60e-1(1.41e-2)+	9.65e-1(9.89e-3)≈	9.65e-1(1.00e-2)≈	9.71e-1(2.23e-2)
Arrhythmia	4.20e-1(6.69e-2)+	4.09e-1(5.04e-2)+	4.80e-1(5.93e-2)≈	5.07e-1(4.27e-2)≈	4.67e-1(5.34e-2)+	4.94e-1(4.54e-2)≈	5.08e-1(6.47e-2)
LSVT	7.65e-1(5.24e-2)+	7.81e-1(5.05e-2)+	8.58e-1(4.33e-2)+	8.68e-1(5.58e-2)≈	8.57e-1(6.65e-2)≈	8.84e-1(3.88e-2)≈	8.87e-1(4.46e-2)
Madelon	5.91e-1(2.04e-2)+	6.42e-1(3.47e-2)+	8.19e-1(2.36e-2)+	9.09e-1(6.20e-3)≈	9.07e-1(5.70e-3)≈	8.78e-1(1.35e-2)+	9.09e-1(6.39e-3)
Gametes	4.01e-1(1.27e-2)+	3.81e-1(1.45e-2)+	5.10e-1(1.51e-2)+	5.73e-1(1.32e-1)+	6.23e-1(6.76e-2)+	6.40e-1(1.54e-1)+	7.94e-1(1.04e-1)
ORL	5.64e-1(2.53e-2)+	6.03e-1(3.73e-2)+	7.14e-1(2.55e-2)+	7.89e-1(2.51e-2)+	7.85e-1(3.13e-2)+	8.06e-1(2.39e-2)≈	8.06e-1(2.32e-2)
SRBCT	5.77e-1(4.00e-2)+	6.17e-1(4.85e-2)+	7.54e-1(4.88e-2)+	9.33e-1(4.33e-2)≈	9.31e-1(3.88e-2)≈	9.45e-1(4.03e-2)≈	9.52e-1(4.13e-2)
DLBCL	5.20e-1(5.64e-2)+	5.34e-1(5.24e-2)+	6.99e-1(7.16e-2)+	9.10e-1(1.07e-1)≈	8.61e-1(9.34e-2)≈	8.76e-1(8.80e-2)≈	9.13e-1(9.37e-2)
Prostate-GE	3.15e-1(1.92e-2)+	3.28e-1(2.10e-2)+	4.20e-1(2.60e-2)+	4.26e-1(1.74e-1)+	5.16e-1(3.68e-2)≈	5.15e-1(3.75e-2)≈	5.32e-1(3.01e-2)
ALLAML	2.87e-1(2.52e-2)+	2.97e-1(2.53e-2)+	3.93e-1(2.89e-2)+	5.15e-1(2.22e-2)+	5.35e-1(1.82e-2)≈	5.27e-1(2.36e-2)≈	5.27e-1(3.46e-2)
+/ − / ≈	**15/0/3**	**17/0/1**	**10/0/8**	**5/0/13**	**4/0/14**	**2/0/16**	–
Rank	6.17	6.33	4.39	2.67	3.83	2.56	**2.06**

TABLE IV
COMPARATIVE RESULTS OF **IGD** VALUES (MEAN AND STANDARD DEVIATION) OBTAINED BY NSGA-II, MOEA/D, SIOM-NSGA-II, SPARSEEA, PMMOEA, DAEA, AND PRDH ON THE TEST DATA

Dataset	**NSGA-II** [49]	**MOEA/D** [51]	**SIOM-NSGA-II** [44]	**SparseEA** [30]	**PMMOEA** [31]	**DAEA** [41]	**PRDH**
Vehicle	5.76e-2(1.49e-2)≈	8.46e-2(1.79e-2)+	5.94e-2(2.00e-2)≈	5.46e-2(1.15e-2)≈	5.76e-2(1.61e-2)≈	5.78e-2(1.50e-2)≈	5.66e-2(1.25e-2)
German	1.25e-1(5.78e-2)≈	1.24e-1(5.41e-2)≈	1.26e-1(5.14e-2)≈	1.22e-1(6.26e-2)≈	1.09e-1(3.31e-2)≈	1.18e-1(4.60e-2)≈	1.12e-1(4.19e-2)
Ionosphere	4.03e-2(1.67e-2)≈	4.46e-2(1.71e-2)≈	4.36e-2(1.81e-2)≈	3.85e-2(1.77e-2)≈	4.49e-2(1.63e-2)≈	3.83e-2(1.56e-2)≈	4.05e-2(1.74e-2)
Sonar	8.29e-2(2.05e-2)+	8.32e-2(2.04e-2)+	7.82e-2(1.55e-2)≈	7.09e-2(1.28e-2)≈	7.66e-2(1.89e-2)≈	7.46e-2(1.27e-2)≈	7.08e-2(1.60e-2)
Mice	5.93e-2(2.31e-2)+	9.05e-2(9.13e-3)+	2.72e-2(5.31e-3)≈	2.39e-2(4.55e-3)−	2.60e-2(5.68e-3)≈	2.53e-2(5.15e-3)≈	2.94e-2(1.05e-2)
Movementlibras	1.18e-1(2.28e-2)+	1.26e-1(1.84e-2)+	8.18e-2(1.93e-2)≈	7.16e-2(1.88e-2)≈	8.23e-2(1.84e-2)≈	7.65e-2(2.01e-2)≈	7.77e-2(2.03e-2)
Hillvalley	4.83e-2(1.67e-2)+	4.14e-2(1.24e-2)+	3.87e-2(9.82e-3)≈	4.04e-2(1.12e-2)≈	4.50e-2(1.43e-2)+	3.53e-2(9.66e-3)≈	3.47e-2(9.77e-3)
Musk1	7.97e-2(2.05e-2)+	7.32e-2(1.75e-2)+	5.46e-2(1.81e-2)≈	5.16e-2(1.48e-2)≈	6.46e-2(2.22e-2)≈	5.38e-2(1.35e-2)≈	5.46e-2(1.95e-2)
Semeion	1.83e-1(1.24e-2)+	1.09e-1(1.19e-2)+	9.14e-2(1.14e-2)+	5.33e-2(1.14e-2)+	4.95e-2(1.44e-2)≈	6.50e-2(1.82e-2)+	4.41e-2(1.91e-2)
Arrhythmia	1.35e-1(5.26e-2)+	1.47e-1(3.63e-2)+	8.34e-2(3.99e-2)≈	6.44e-2(2.30e-2)≈	9.16e-2(3.73e-2)+	7.76e-2(2.58e-2)≈	7.19e-2(3.67e-2)
LSVT	1.46e-1(3.36e-2)+	1.27e-1(3.46e-2)+	7.96e-2(2.71e-2)+	7.05e-2(4.03e-2)≈	8.15e-2(5.48e-2)≈	6.37e-2(2.45e-2)≈	6.12e-2(2.40e-2)
Madelon	2.52e-1(1.58e-2)+	1.99e-1(2.41e-2)+	1.07e-1(1.30e-2)+	1.60e-2(5.58e-3)−	2.00e-2(4.46e-3)≈	9.46e-2(2.23e-2)+	2.03e-2(4.91e-3)
Gametes	4.30e-1(1.63e-2)+	4.57e-1(1.93e-2)+	2.96e-1(1.69e-2)+	2.66e-1(1.43e-1)+	2.12e-1(6.52e-2)+	2.06e-1(1.54e-1)+	7.49e-2(9.12e-2)
ORL	3.20e-1(9.82e-3)+	2.70e-1(1.84e-2)+	1.84e-1(1.00e-2)+	6.26e-2(1.32e-2)≈	6.48e-2(1.45e-2)+	5.56e-2(1.09e-2)≈	5.71e-2(8.57e-3)
SRBCT	3.77e-1(1.95e-2)+	3.22e-1(2.92e-2)+	1.88e-1(2.81e-2)+	4.94e-2(2.40e-2)≈	4.65e-2(2.41e-2)≈	4.68e-2(2.06e-2)≈	4.06e-2(2.06e-2)
DLBCL	4.56e-1(3.18e-2)+	4.27e-1(3.74e-2)+	2.60e-1(5.43e-2)+	9.19e-2(1.14e-1)≈	1.44e-1(9.81e-2)≈	1.25e-1(9.62e-2)≈	9.15e-2(9.47e-2)
Prostate-GE	4.17e-1(6.78e-3)+	3.87e-1(1.11e-2)+	1.93e-1(9.09e-3)+	1.31e-1(1.91e-1)+	3.29e-2(4.04e-2)≈	3.33e-2(4.12e-2)≈	1.49e-2(3.30e-2)
ALLAML	4.31e-1(1.14e-2)+	4.07e-1(1.18e-2)+	2.10e-1(1.56e-2)+	3.36e-2(2.44e-2)≈	1.13e-2(1.99e-2)≈	2.06e-2(2.59e-2)≈	2.00e-2(2.38e-2)
+/ − / ≈	**15/0/3**	**16/0/2**	**9/0/9**	**3/2/13**	**4/0/14**	**3/0/15**	–
Rank	6.28	6.22	4.78	2.39	3.56	2.61	**2.17**

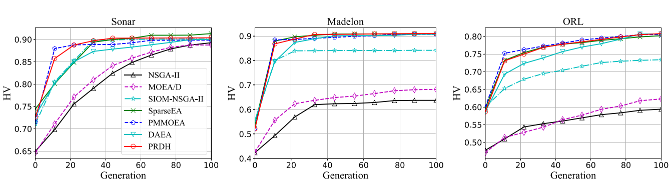

Fig. 5. Convergence curves obtained by each algorithm on training sets in terms of HV.

classification error rate and the duplication handling technique can improve the diversity of the solution set.

To show the performance of different algorithms under different computational resources, Fig. 5 plots the average convergence curves obtained by each algorithm on training sets in terms of HV. As can be seen from the figure, except for PMMOEA, PRDH can quickly converge to a stable HV value, which is also a large HV compared to most other algorithms. By contrast, although DAEA and PRDH adopt a similar initialization strategy, DAEA requires more iterations to converge to a steady HV.

In terms of the IGD results in Table IV, it can be observed that no method can perform the best on all the datasets. But for the proposed PRDH method, it has the best performance on most datasets. Specifically, PRDH wins on 8 out of 18 test datasets, while SparseEA, DAEA, and SIOM-NSGA-II win on six, two, and two datasets, respectively, out of 18 test datasets. The rest three algorithms: 1) NSGA-II; 2) MOEA/D; and 3) SIOM-NSGA-II perform the best on none of the 18 datasets in terms of the IGD value. In terms of Friedman's rank results in Table IV, the proposed PRDH method can also have the best ranking among all the compared MOEAs.

B. Comparisons in Terms of the Classification Performance

From the perspective of decision makers, classification performance is more important than the number of selected features. Table V summarizes the comparison results derived from seven MOEAs in terms of the MCER metric. We can observe that the classifier employing PRDH as a feature selection method leads to superior classification performance. On the low-dimensional datasets, SparseEA usually exhibits the best MCER values. On the medium and high-dimensional datasets, the proposed PRDH has the best classification performance, as it gains the best MCER results on most

TABLE V
COMPARATIVE RESULTS OF **MCER** VALUES (MEAN AND STANDARD DEVIATION) OBTAINED BY NSGA-II, MOEA/D, SIOM-NSGA-II, SPARSEEA, PMMOEA, DAEA, AND PRDH ON THE TEST DATA

Dataset	**NSGA-II** [49]	**MOEA/D** [51]	**SIOM-NSGA-II** [44]	**SparseEA** [30]	**PMMOEA** [31]	**DAEA** [41]	**PRDH**
Vehicle	28.7%(2.09e-2)≈	31.6%(2.42e-2)+	28.7%(2.16e-2)≈	28.5%(1.42e-2)≈	28.8%(1.80e-2)≈	28.9%(1.76e-2)≈	29.0%(1.72e-2)
German	33.7%(7.59e-2)≈	35.1%(5.97e-2)+	34.5%(6.30e-2)+	33.2%(7.50e-2)≈	31.9%(3.18e-2)≈	32.0%(6.22e-2)≈	31.7%(5.81e-2)
Ionosphere	6.55%(2.16e-2)≈	7.44%(2.40e-2)≈	6.95%(2.51e-2)≈	6.37%(2.38e-2)≈	7.33%(2.26e-2)≈	6.08%(2.05e-2)≈	6.59%(2.16e-2)
Sonar	17.9%(4.08e-2)≈	18.6%(3.11e-2)≈	18.2%(3.27e-2)≈	17.1%(2.81e-2)≈	18.6%(3.13e-2)≈	18.1%(2.28e-2)≈	17.8%(2.64e-2)
Mice	1.48%(8.74e-3)≈	2.72%(1.12e-2)+	1.82%(1.06e-2)+	1.39%(7.51e-3)≈	1.57%(7.42e-3)≈	1.46%(8.27e-3)≈	1.68%(9.80e-3)
Movementlibras	23.9%(4.06e-2)≈	24.5%(3.38e-2)≈	24.9%(3.74e-2)≈	23.4%(2.93e-2)≈	25.1%(3.22e-2)≈	24.5%(3.17e-2)≈	24.2%(3.70e-2)
Hillvalley	40.3%(2.00e-2)≈	41.2%(1.90e-2)+	40.5%(1.51e-2)≈	40.9%(1.53e-2)≈	41.9%(2.05e-2)+	40.3%(1.57e-2)≈	40.2%(1.69e-2)
Musk1	12.8%(2.92e-2)≈	16.0%(3.14e-2)+	13.9%(3.05e-2)≈	13.7%(2.61e-2)≈	15.6%(3.58e-2)≈	12.7%(2.77e-2)≈	13.8%(3.05e-2)
Semeion	2.07%(8.95e-3)≈	6.01%(2.62e-2)+	2.33%(1.03e-2)≈	3.20%(1.50e-2)+	2.81%(1.21e-2)+	1.78%(8.66e-3)≈	2.13%(2.34e-2)
Arrhythmia	61.4%(7.30e-2)+	62.3%(5.78e-2)+	56.7%(6.62e-2)≈	53.7%(4.80e-2)≈	58.3%(5.96e-2)+	55.1%(5.05e-2)≈	53.6%(7.24e-2)
LSVT	16.9%(5.52e-2)+	18.5%(5.18e-2)+	14.3%(4.61e-2)≈	14.1%(6.18e-2)≈	15.3%(7.37e-2)≈	12.2%(4.31e-2)≈	11.9%(4.98e-2)
Madelon	30.6%(1.98e-2)+	29.8%(2.72e-2)+	16.4%(1.97e-2)+	9.63%(6.87e-3)≈	9.88%(6.34e-3)≈	11.2%(1.18e-2)+	9.65%(7.16e-3)
Gametes	47.6%(1.86e-2)+	48.2%(2.30e-2)+	46.7%(1.63e-2)+	46.9%(1.46e-1)+	41.4%(7.43e-2)+	39.5%(1.69e-1)+	22.5%(1.14e-1)
ORL	23.1%(3.71e-2)+	24.5%(4.36e-2)+	21.3%(2.81e-2)≈	22.7%(2.84e-2)+	23.2%(3.52e-2)+	20.8%(2.68e-2)≈	20.8%(2.64e-2)
SRBCT	16.4%(6.46e-2)+	17.6%(6.95e-2)+	13.7%(6.33e-2)+	7.31%(4.77e-2)≈	7.49%(4.28e-2)+	5.97%(4.44e-2)≈	5.22%(4.55e-2)
DLBCL	19.4%(9.97e-2)+	21.8%(7.85e-2)+	17.8%(9.41e-2)+	9.93%(1.18e-1)≈	15.3%(1.03e-1)+	13.7%(9.69e-2)≈	9.51%(1.03e-1)
Prostate-GE	54.6%(3.49e-2)+	54.7%(3.75e-2)+	54.5%(3.45e-2)+	63.1%(1.91e-1)+	53.3%(4.05e-2)+	53.3%(4.13e-2)+	51.5%(3.31e-2)
ALLAML	59.0%(4.37e-2)+	59.1%(4.47e-2)+	57.7%(3.85e-2)+	53.4%(2.44e-2)+	51.1%(2.00e-2)≈	52.1%(2.60e-2)≈	52.0%(3.80e-2)
$+/-/\approx$	**9/0/9**	**15/0/3**	**7/0/11**	**5/0/13**	**8/0/10**	**3/0/15**	–
Rank	4.39	6.67	4.56	3.06	4.50	2.61	2.22

TABLE VI
COMPARATIVE RESULTS OF **FN** VALUES (MEAN AND STANDARD DEVIATION) OBTAINED BY NSGA-II, MOEA/D, SIOM-NSGA-II, SPARSEEA, PMMOEA, DAEA, AND PRDH ON THE TEST DATA

Dataset	**NSGA-II** [49]	**MOEA/D** [51]	**SIOM-NSGA-II** [44]	**SparseEA** [30]	**PMMOEA** [31]	**DAEA** [41]	**PRDH**
Vehicle	6.97e+0(1.7e+0)≈	4.80e+0(7.6e-1)−	6.83e+0(2.2e+0)≈	7.97e+0(1.7e+0)+	6.97e+0(1.7e+0)≈	6.90e+0(1.6e+0)≈	6.77e+0(1.2e+0)
German	4.53e+0(3.2e+0)≈	3.97e+0(1.7e+0)≈	4.27e+0(2.9e+0)≈	4.43e+0(2.3e+0)≈	4.70e+0(2.5e+0)≈	4.37e+0(2.7e+0)≈	4.77e+0(2.5e+0)
Ionosphere	4.07e+0(1.5e+0)≈	3.40e+0(1.3e+0)≈	3.87e+0(1.4e+0)≈	3.73e+0(1.5e+0)≈	3.73e+0(1.6e+0)≈	3.80e+0(1.2e+0)≈	4.03e+0(1.3e+0)
Sonar	1.06e+1(4.3e+0)+	7.67e+0(3.5e+0)≈	7.30e+0(3.5e+0)≈	9.93e+0(5.2e+0)≈	6.80e+0(3.4e+0)≈	9.60e+0(5.1e+0)≈	7.80e+0(2.8e+0)
Mice	1.92e+1(3.8e+0)≈	1.25e+1(1.5e+0)−	1.58e+1(2.7e+0)−	1.81e+1(4.8e+0)≈	1.81e+1(3.8e+0)≈	1.91e+1(4.6e+0)≈	1.79e+1(4.0e+0)
Movementlibras	1.51e+1(6.8e+0)+	1.07e+1(2.8e+0)≈	1.09e+1(5.6e+0)≈	1.25e+1(6.4e+0)≈	1.00e+1(4.2e+0)≈	1.29e+1(6.6e+0)≈	1.01e+1(3.5e+0)
Hillvalley	1.22e+1(5.5e+0)+	5.53e+0(2.5e+0)−	8.37e+0(4.1e+0)≈	7.13e+0(3.6e+0)≈	4.63e+0(2.0e+0)−	8.83e+0(4.4e+0)≈	7.13e+0(2.2e+0)
Musk1	3.24e+1(1.4e+1)+	1.53e+1(4.7e+0)≈	2.13e+1(9.3e+0)≈	2.08e+1(1.4e+1)≈	1.62e+1(8.6e+0)≈	2.87e+1(1.1e+1)+	1.75e+1(9.2e+0)
Semeion	5.83e+1(1.4e+1)+	2.20e+1(4.4e+0)≈	3.88e+1(1.3e+1)+	6.01e+1(2.8e+1)+	3.41e+1(1.8e+1)+	5.30e+1(2.0e+1)+	2.12e+1(1.9e+1)
Arrhythmia	2.41e+1(1.2e+1)+	2.40e+1(8.7e+0)+	1.09e+1(6.2e+0)≈	8.87e+0(3.7e+0)≈	1.00e+1(4.9e+0)≈	1.28e+1(7.3e+0)≈	1.05e+1(4.6e+0)
LSVT	3.71e+1(9.4e+0)+	2.30e+1(6.7e+0)+	1.18e+1(7.3e+0)+	5.80e+0(2.8e+0)≈	5.03e+0(2.3e+0)≈	1.22e+1(9.7e+0)+	6.70e+0(4.7e+0)
Madelon	1.07e+2(9.1e+0)+	7.06e+1(1.4e+1)+	2.47e+1(6.0e+0)+	7.07e+0(1.7e+0)≈	6.30e+0(1.1e+0)≈	1.68e+1(3.1e+0)+	7.33e+0(2.1e+0)
Gametes	3.52e+2(3.2e+1)+	3.69e+2(2.5e+1)+	1.47e+2(2.6e+1)+	1.20e+0(6.1e-1)−	1.75e+1(6.6e+1)+	3.03e+1(8.2e+1)≈	7.77e+0(2.1e+1)
ORL	3.43e+2(2.3e+1)+	2.61e+2(2.8e+1)+	1.55e+2(3.3e+1)+	4.56e+1(4.7e+1)≈	4.11e+1(2.0e+1)≈	3.89e+1(1.9e+1)≈	4.17e+1(3.1e+1)
SRBCT	8.26e+2(3.1e+1)+	6.72e+2(3.9e+1)+	3.63e+2(2.3e+1)+	3.93e+0(1.3e+0)−	5.23e+0(2.4e+0)≈	4.30e+0(1.4e+0)−	6.97e+0(3.4e+0)
DLBCL	2.23e+3(4.3e+1)+	2.01e+3(6.7e+1)+	1.01e+3(5.3e+1)+	2.13e+0(1.1e+0)−	3.63e+0(2.2e+0)−	2.67e+0(1.1e+0)−	4.53e+0(2.0e+0)
Prostate-GE	2.47e+3(3.3e+1)+	2.28e+3(7.3e+1)+	1.11e+3(2.5e+1)+	2.30e+0(2.4e+0)≈	1.97e+0(1.1e+0)≈	1.93e+0(1.2e+0)≈	2.53e+0(2.4e+0)
ALLAML	3.00e+3(6.6e+1)+	2.81e+3(8.0e+1)+	1.38e+3(3.6e+1)+	1.20e+0(4.1e-1)−	1.57e+0(5.0e-1)−	1.37e+0(4.9e-1)−	2.37e+0(1.7e+0)
$+/-/\approx$	**14/0/4**	**9/3/6**	**9/1/8**	**2/4/12**	**2/3/13**	**4/3/11**	–
Rank	6.69	3.83	4.17	3.33	2.56	3.94	3.47

datasets. This is due to the fact that among the two objectives in feature selection, the proposed problem reformulation and constraint-handling methods can focus more on the improvement of classification performance. Wilcoxon's rank-sum test results show that PRDH is significantly better than its six competitors (i.e., NSGA-II, MOEA/D, SIOM-NSGA-II, SparseEA, PMMOEA, and DAEA) on 9, 15, 7, 5, 8, and 3 out of the 18 datasets, respectively, and is not significantly worse than the compared algorithms on any dataset. Besides, Friedman's rank results indicate that PRDH achieves the first ranking, which suggests that it has the best performance in terms of the MCER metric.

C. Comparisons in Terms of the Number of Selected Features

Table VI presents the detailed comparison results in terms of the FN metric among the seven MOEAs. It can be seen that MOEA/D achieves the best FN results on the low-dimensional datasets, while SparseEA exhibits the best FN results on the high-dimensional datasets. In terms of Friedman's rank test, PMMOEA ranks first, followed by SparseEA and PRDH.

In general, PRDH selects a smaller number of features than NSGA-II, MOEA/D, and SIOM-NSGA-II on the high-dimensional datasets, but selects more features than PMMOEA, SparseEA, and DAEA. The reasons can be attributed that are listed in the following.

1) The classification performance and the number of selected features are often in conflict with each other. The MCER results in Table V suggest that PRDH has the best overall classification performance on high-dimensional datasets, so it might cause PRDH to select more features on high-dimensional datasets than PMMOEA, SparseEA, and DAEA.
2) The proposed problem reformulation method divides the whole population into feasible solutions and infeasible solutions based on their classification performance, and the constraint-handling technique used in PRDH prefers to select feasible solutions which often select a larger number of features but exhibit better classification performance. The infeasible solutions that select a smaller number of features with poor classification performance tend to be discarded.

However, as discussed before, in feature selection, classification performance is more important than the number of selected features. It is acceptable to sacrifice increasing a

TABLE VII
COMPARATIVE RESULTS OF **TRAINING TIME** VALUES (MEAN AND STANDARD DEVIATION) OBTAINED BY NSGA-II, MOEA/D, SIOM-NSGA-II, SPARSEEA, PMMOEA, DAEA, AND PRDH

Dataset	**NSGA-II** [49]	**MOEA/D** [51]	**SIOM-NSGA-II** [44]	**SparseEA** [30]	**PMMOEA** [31]	**DAEA** [41]	**PRDH**
Vehicle	2.47e+1(1.6e+0)≈	2.46e+1(6.6e-1)+	2.53e+1(1.4e+0)+	2.53e+1(1.6e+0)+	2.92e+1(2.1e+0)+	2.65e+1(1.5e+0)+	2.43e+1(5.5e-1)
German	4.38e+1(3.5e+0)−	4.65e+1(1.9e+0)≈	4.07e+1(3.3e+0)−	4.54e+1(1.9e+0)−	5.15e+1(3.3e+0)+	4.82e+1(1.3e+0)+	4.72e+1(2.3e+0)
Ionosphere	8.84e+0(1.3e+0)+	9.13e+0(3.6e-1)+	1.25e+1(1.1e+0)+	1.30e+1(1.1e+0)+	1.49e+1(2.6e+0)+	1.34e+1(8.6e-1)+	8.71e+0(1.4e-1)
Sonar	8.98e+0(1.2e+0)≈	9.58e+0(7.8e-1)+	2.17e+1(1.9e+0)+	2.28e+1(2.6e+0)+	1.37e+1(2.3e+0)+	2.39e+1(3.0e+0)+	8.15e+0(3.5e-1)
Mice	1.92e+2(8.1e+0)+	1.79e+2(4.7e+0)≈	1.94e+2(1.0e+1)+	2.04e+2(1.6e+1)+	1.97e+2(1.5e+1)+	2.17e+2(1.9e+1)+	1.77e+2(2.8e+0)
Movementlibras	3.82e+1(5.3e+0)+	3.57e+1(4.4e+0)+	6.13e+1(4.9e+0)+	6.43e+1(6.8e+0)+	3.68e+1(4.7e+0)+	6.73e+1(6.8e+0)+	3.06e+1(8.2e-1)
Hillvalley	3.41e+2(4.3e+1)+	2.90e+2(1.7e+1)+	2.99e+2(1.5e+1)+	3.07e+2(2.0e+1)+	2.77e+2(2.1e+1)−	3.47e+2(1.4e+1)+	2.78e+2(4.1e+0)
Musk1	1.21e+2(4.6e+0)+	1.07e+2(3.1e+0)+	1.97e+2(5.4e+0)+	1.98e+2(1.3e+1)+	9.80e+1(9.7e+0)+	2.25e+2(1.3e+1)+	9.16e+1(2.6e+0)
Semeion	2.29e+3(1.1e+2)+	1.66e+3(7.7e+1)+	1.89e+3(7.4e+1)+	1.73e+3(1.3e+2)+	1.55e+3(1.3e+2)+	2.20e+3(6.9e+1)+	1.43e+3(1.2e+2)
Arrhythmia	1.42e+2(1.6e+1)+	1.46e+2(1.3e+1)+	2.76e+2(1.1e+1)+	2.65e+2(6.5e+0)+	1.24e+2(4.7e+0)≈	3.13e+2(1.5e+1)+	1.23e+2(2.3e+0)
LSVT	2.95e+1(9.8e-1)+	3.17e+1(1.4e+0)+	1.83e+2(3.5e+0)+	1.77e+2(3.0e+0)+	2.70e+1(1.8e+0)+	1.94e+2(3.3e+0)+	2.27e+1(5.3e-1)
Madelon	9.96e+3(2.6e+2)+	8.90e+3(4.6e+2)+	5.93e+3(1.4e+2)+	3.78e+3(1.8e+2)−	3.55e+3(2.9e+2)−	8.09e+3(1.4e+2)+	4.06e+3(3.7e+2)
Gametes	8.37e+3(2.4e+2)+	8.83e+3(2.7e+2)+	4.71e+3(2.3e+2)+	3.27e+3(2.0e+3)≈	2.76e+3(2.6e+3)≈	2.77e+3(1.1e+3)+	1.94e+3(3.9e+2)
ORL	6.70e+2(8.8e+1)+	5.90e+2(2.5e+1)+	6.06e+2(8.4e+1)+	3.94e+2(5.3e+1)+	2.30e+2(1.4e+1)≈	3.91e+2(1.1e+1)+	3.00e+2(1.5e+2)
SRBCT	1.63e+2(7.1e+0)+	1.57e+2(7.5e+0)+	3.57e+2(6.3e+0)+	2.64e+2(7.8e+0)+	1.15e+2(7.5e+0)+	3.14e+2(4.8e+0)+	1.14e+2(4.3e+1)
DLBCL	3.99e+2(3.9e+1)+	3.59e+2(8.2e+0)+	6.57e+2(2.7e+1)+	4.24e+2(3.2e+1)+	3.09e+2(5.1e+1)+	5.65e+2(6.0e+1)+	2.38e+2(6.6e+0)
Prostate-GE	5.77e+2(1.5e+1)+	5.54e+2(1.3e+1)+	9.23e+2(1.4e+2)+	5.63e+2(8.9e+1)+	4.14e+2(1.9e+1)−	7.31e+2(1.1e+2)+	4.16e+2(1.2e+2)
ALLAML	5.21e+2(7.7e+1)+	4.52e+2(1.2e+1)+	8.69e+2(1.2e+2)+	2.81e+2(6.6e+0)−	4.57e+2(1.4e+2)+	8.26e+2(2.3e+2)+	3.36e+2(3.7e+1)
+/ − / ≈	**15/1/2**	**16/0/2**	**17/1/0**	**14/3/1**	**12/3/3**	**18/0/0**	–
Rank	4.28	3.50	5.17	4.44	3.06	6.00	**1.56**

certain number of selected features to improve classification performance.

D. Comparisons in Terms of the Training Time

Training time is an important indicator to verify the efficiency of an algorithm for feature selection problems. Table VII presents the average training time achieved by PRDH and the six compared MOEAs on the 18 classification datasets. From the table, we can observe that the proposed PRDH method consumes the least training time on most of the datasets. Although both DAEA and the proposed PRDH method involve duplication handling, the training time of DAEA is much longer than that of PRDH. This is because during the environmental selection process, DAEA calculates the distance between every two solutions in the population, which leads to the computational complexity of $O(N^2D)$, which is very time consuming. By contrast, for the proposed PRDH method, it only calculates the distance between the duplicated solution and two nondominated solutions, respectively, with the same or similar objective value on each of the two objectives with that of the duplicated solution, which only leads to the computational complexity of $\max\{O(N^2), O(ND)\}$. Among these seven MOEAs, the proposed PRDH method consumes relatively less training time as it gets the first rank according to Friedman's rank results, which verifies its efficiency in solving multiobjective feature selection problems.

VI. FURTHER ANALYSIS

A. Effectiveness of the Problem Reformulation and Constraint Handling Techniques

In the proposed PRDH algorithm, the problem reformulation and constraint handling techniques work together to contribute to improving the classification performance of PRDH. To investigate their effectiveness, we compare the performance of PRDH and PRDH without the proposed problem reformulation and constraint handling mechanisms, abbreviated as PRDH-woPR. Without the proposed problem reformulation and constraint handling mechanisms, PRDH-woPR adopts the nondominated sorting, duplication handling, and crowding distance operators to select elite solutions as the next parent population.

TABLE VIII
COMPARATIVE RESULTS OF **HV** AND **MCER** VALUES (MEAN AND STANDARD DEVIATION) OBTAINED BY PRDH-WOPR AND PRDH ON THE TEST DATA, RESPECTIVELY

Dataset	**HV**		**MCER**	
	PRDH-woPR	**PRDH**	**PRDH-woPR**	**PRDH**
Vehicle	6.79e-1(1.13e-2)	6.77e-1(1.31e-2)	28.7%(1.65e-2)	29.0%(1.72e-2)
German	6.61e-1(5.72e-2)	6.75e-1(4.62e-2)	33.7%(6.85e-2)	31.7%(5.81e-2)
Ionosphere	9.10e-1(2.34e-2)	9.11e-1(1.92e-2)	6.90%(2.68e-2)	6.59%(2.16e-2)
Sonar	8.10e-1(2.94e-2)	8.19e-1(2.18e-2)	18.9%(3.72e-2)	17.8%(2.64e-2)
Mice	9.54e-1(8.72e-3)	9.55e-1(9.55e-3)	1.84%(1.05e-2)	1.68%(9.80e-3)
Movementlibras	7.47e-1(3.09e-2)	7.57e-1(3.13e-2)	25.4%(3.63e-2)	24.2%(3.70e-2)
Hillvalley	6.23e-1(1.57e-2)	6.27e-1(1.46e-2)	40.7%(1.80e-2)	40.2%(1.69e-2)
Musk1	8.56e-1(2.82e-2)	8.63e-1(2.61e-2)	14.7%(3.31e-2)	13.8%(3.05e-2)
Semeion	9.79e-1(1.46e-2)	9.71e-1(2.23e-2)	1.24%(1.52e-2)	2.13%(2.34e-2)
Arrhythmia	4.82e-1(4.55e-2)	5.08e-1(6.47e-2)	56.6%(5.09e-2)	53.6%(7.24e-2)
LSVT	8.65e-1(4.50e-2)	8.87e-1(4.46e-2)	14.4%(4.98e-2)	11.9%(4.98e-2)
Madelon	9.06e-1(7.39e-3)	9.09e-1(6.39e-3)	10.0%(8.22e-3)	9.65%(7.16e-3)
Gametes	7.74e-1(1.12e-1)	7.94e-1(1.04e-1)	24.7%(1.23e-1)	22.5%(1.14e-1)
ORL	7.92e-1(3.15e-2)	8.06e-1(2.32e-2)	22.3%(3.58e-2)	20.8%(2.64e-2)
SRBCT	9.33e-1(5.31e-2)	9.52e-1(4.13e-2)	7.32%(5.85e-2)	5.22%(4.55e-2)
DLBCL	8.64e-1(8.95e-2)	9.13e-1(9.37e-2)	15.0%(9.85e-2)	9.51%(1.03e-1)
Prostate-GE	5.26e-1(2.38e-2)	5.32e-1(3.01e-2)	52.1%(2.62e-2)	51.5%(3.31e-2)
ALLAML	5.36e-1(1.89e-2)	5.27e-1(3.46e-2)	51.1%(2.07e-2)	52.0%(3.80e-2)

Table VIII summarizes the average HV and MCER results obtained by PRDH-woPR and PRDH on the test data over the 30 runs, respectively. As can be seen from the table, PRDH outperforms PRDH-woPR on most of the datasets, which validates the effectiveness of the proposed problem reformulation and constraint handling techniques. Fig. 6 presents the distributions of nondominated feature subsets in terms of median HV value obtained by PRDH-woPR and PRDH on the Sonar, Madelon, and ORL datasets, respectively. On the Sonar dataset, PRDH can obtain more feature subsets with a smaller classification error rate than that of PRDH-woPR. On the Madelon and ORL datasets, under the same or a similar number of selected features, the classification error rates obtained by PRDH tend to be smaller than that of PRDH-woPR, which also verifies the effectiveness of the proposed problem reformulation and constraint handling techniques in improving classification performance.

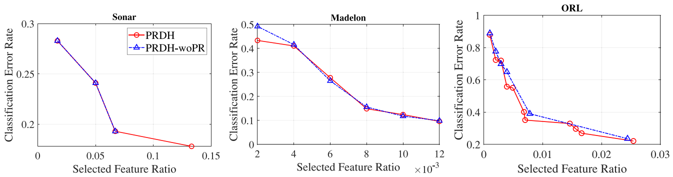

Fig. 6. Distributions of nondominated feature subsets obtained by PRDH and PRDH-woPR on test sets in terms of median HV value.

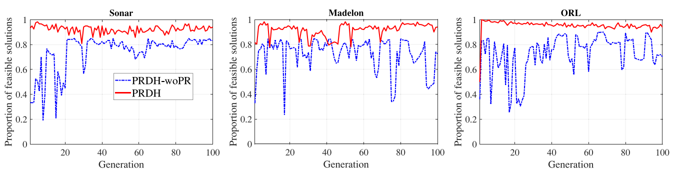

Fig. 7. Proportion of feasible solutions in the population.

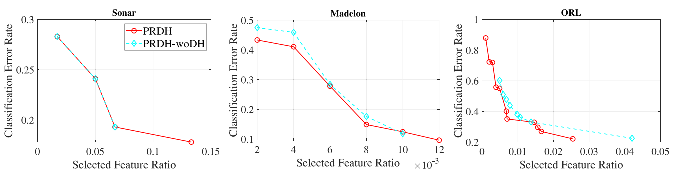

Fig. 8. Distributions of nondominated feature subsets obtained by PRDH and PRDH-woDH on test sets in terms of median HV value.

To further explore the proportion of feasible solutions in the whole population with and without the constraint handling technique, Fig. 7 also plots the proportion of feasible feature subsets to the whole population after problem reformulation. From the figure, it can be observed that without the constraint handling mechanism, the number of feasible solutions obtained by PRDH-woPR is smaller than that of PRDH, which means PRDH-woPR preserves more infeasible feature subsets with higher classification error rates. The proposed problem reformulation and constraint handling mechanisms prefer to select feasible solutions with smaller classification error rates, which can focus more on classification performance. This can explain why the performance of PRDH equipped with the problem reformulation and constraint handling mechanisms can be improved a lot.

B. Effectiveness of the Duplication Handling Method

To verify the effectiveness of the proposed duplication handling mechanism, we perform a comparison on PRDH with and without the proposed duplication handling mechanism (in short called PRDH-woDH), respectively.

The average HV and MCER results obtained by PRDH-woDH and PRDH on the test data over the 30 runs are listed in Table IX. It can be observed that PRDH is better than PRDH-woDH on most of the datasets, which demonstrates that the duplication handling method works well in the PRDH algorithm. In terms of the distributions of nondominated feature subsets presented in Fig. 8, we can see that under the similar number of selected features, the feature subsets obtained by PRDH have smaller classification error rates than that of PRDH-woDH, or under the similar classification error rates,

TABLE IX

COMPARATIVE RESULTS OF **HV** AND **MCER** VALUES (MEAN AND STANDARD DEVIATION) OBTAINED BY PRDH-WODH AND PRDH ON THE TEST DATA, RESPECTIVELY

Dataset	HV		MCER	
	PRDH-woDH	PRDH	PRDH-woDH	PRDH
Vehicle	6.78e-1(1.08e-2)	6.77e-1(1.31e-2)	28.9%(1.63e-2)	29.0%(1.72e-2)
German	6.64e-1(6.57e-2)	6.75e-1(4.62e-2)	33.1%(7.91e-2)	31.7%(5.81e-2)
Ionosphere	9.10e-1(2.20e-2)	9.11e-1(1.92e-2)	6.93%(2.51e-2)	6.59%(2.16e-2)
Sonar	8.07e-1(2.78e-2)	8.19e-1(2.18e-2)	19.2%(3.34e-2)	17.8%(2.64e-2)
Mice	9.55e-1(8.44e-3)	9.55e-1(9.55e-3)	1.79%(1.02e-2)	1.68%(9.80e-3)
Movementlibras	7.50e-1(2.92e-2)	7.57e-1(3.13e-2)	25.0%(3.45e-2)	24.2%(3.70e-2)
Hillvalley	6.22e-1(1.52e-2)	6.27e-1(1.46e-2)	40.8%(1.76e-2)	40.2%(1.69e-2)
Musk1	8.52e-1(2.34e-2)	8.63e-1(2.61e-2)	15.2%(2.79e-2)	13.8%(3.05e-2)
Semeion	9.70e-1(1.97e-2)	9.71e-1(2.23e-2)	2.21%(2.14e-2)	2.13%(2.34e-2)
Arrhythmia	4.75e-1(5.35e-2)	5.08e-1(6.47e-2)	57.3%(5.97e-2)	53.6%(7.24e-2)
LSVT	8.57e-1(4.91e-2)	8.87e-1(4.46e-2)	15.2%(5.45e-2)	11.9%(4.98e-2)
Madelon	9.08e-1(7.36e-3)	9.09e-1(6.39e-3)	9.78%(8.47e-3)	9.65%(7.16e-3)
Gametes	8.05e-1(9.05e-2)	7.94e-1(1.04e-1)	21.3%(9.94e-2)	22.5%(1.14e-1)
ORL	7.91e-1(2.59e-2)	8.06e-1(2.32e-2)	22.4%(2.98e-2)	20.8%(2.64e-2)
SRBCT	9.53e-1(4.13e-2)	9.52e-1(4.13e-2)	5.06%(4.52e-2)	5.22%(4.55e-2)
DLBCL	8.96e-1(8.65e-2)	9.13e-1(9.37e-2)	11.4%(9.52e-2)	9.51%(1.03e-1)
Prostate-GE	5.36e-1(1.76e-2)	5.32e-1(3.01e-2)	51.0%(1.94e-2)	51.5%(3.31e-2)
ALLAML	5.26e-1(3.52e-2)	5.27e-1(3.46e-2)	52.1%(3.87e-2)	52.0%(3.80e-2)

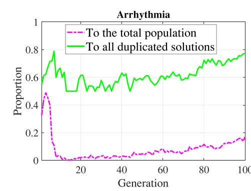

Fig. 9. Proportion of deleted duplicated solutions (in the objective space) in the total population (dashed line) and all duplicated solutions (solid line).

the feature subsets obtained by PRDH select smaller number of features than that of PRDH-woDH. This is because the proposed duplication handling method can remove some duplicated feature subsets that contain more irrelevant and redundant features, which is beneficial to reduce redundancy to some extent and improve classification performance.

Fig. 9 also gives an example that plots the proportions of deleted duplicated solutions by PRDH to the total population size and all duplicated solutions in the population, respectively. As can be seen from the figure, at each generation, a number of duplicated solutions are removed from the population. Meanwhile, the deleted duplicated solutions account for more than half of all duplicated solutions, which remove solutions with a high similarity but exhibit worse classification performance than the nondominated solutions.

C. Threats to Validity

There are some threats to validity. The threat to internal validity comes from the noise of labels. When dealing with real-world datasets, it is common for some instances to be mislabeled. Since the proposed method needs the class labels, the presence of class label noise can affect the results of the selected features, thus inevitably degrading the classification performance. To reduce the negative impact of mislabeled data on the feature selection process, designing a strategy to deal with label noise can improve the robustness of the feature selection method.

The threat to external validity lies in the choice of datasets. Our results show that the datasets used in this article contain a large number of irrelevant and redundant features, and only selecting a small number of informative features from them can significantly improve the classification performance. Thus, the results might not generalize to other real-world applications. To minimize this risk, in this article, we select datasets from different domains, such as medical, DNA microarrays, signal processing, and finance.

VII. Conclusion

In this article, considering the characteristics of the multiobjective feature selection problem, a novel MOEA called PRDH has been proposed for feature selection in classification. In PRDH, a problem reformulation method and a constraint-handling method were designed for handling the different preferences between objectives in multiobjective feature selection. Besides, a duplication handling method based on the distribution of feature subsets in the objective space and their similarity in the search space was proposed to address the frequently appeared duplicated solutions in the multiobjective space.

In terms of several performance metrics, such as HV, IGD, classification performance, the number of selected features, and training time, the effectiveness and efficiency of PRDH have been verified through experiments on 18 classification datasets. Additionally, the roles and contributions of key components of PRDH have also been demonstrated, which suggests that each component can contribute positively to the search and classification performance, and the combination of them achieved the best results.

Although the proposed method is promising, much work can be done to further realize its potential. First, one limitation of PRDH is that it selects a few more features than the state-of-the-art MOEA-based feature selection methods. It is worth removing more redundant and irrelevant features based on the proposed duplication analysis method. Second, in addition to imposing constraints on the classification performance, other methods can also be used to handle different preferences between objectives. For instance, using preference-inspired algorithms or setting customized weight vectors for different objectives. Third, the initialization strategy plays an important role in high-dimensional feature selection. It is desirable to design a new initialization strategy to better cover the whole search and objective space.

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